

2022

Time : 3 hours

Full Marks : 50

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

*Answer from **all** the Sections as directed.*

Section – A

(Objective Type Questions)

1. [A] Fill in the blanks : 1×5 = 5
- (a) The relation between the mean variance of χ^2 with n degree of freedom is _____.
- (b) The range of F-variate is _____.
- (c) There are _____ basic principle of design.

(d) The sample mean is _____ estimate of the population mean.

(e) The probability of committing Type I error is _____.

[B] Choose the correct answer from the given alternatives : $1 \times 5 = 5$

(a) The hypothesis may be classified as :

(i) Simple

(ii) Compositive

☒ (iii) Null

☒ (iv) All of these

(b) Power of the test is related to :

(i) Type I error

☒ (ii) Type II error

☒ (iii) Both (i) and (ii)

(iv) None of these

(c) Whether a test is one-sided or two sided depends on :

☒ (i) Alternative hypothesis

(ii) Compositive hypothesis

(iii) Null hypothesis

(iv) Simple hypothesis

(d) The degree of freedom of error for R. B. D. is :

☒ (i) $(r-1)(t-1)$

(ii) $(t-1)$

(iii) $(r-1)$

☒ (iv) $(rt-1)$

(e) According to Prof. R. A. Fisher, there are _____ criteria of a good estimate.

☒ (i) 4

(ii) 3

(iii) 2

(iv) 1

Section - B

(Short-answer Type Questions)

2. Answer any four questions of the following :

$3 \times 4 = 12$

☒ (a) Discuss the problem of estimator with example.

☒ (b) Define Null and Alternative hypothesis.

☒ (c) Explain Completely Randomised Design (C. R. D).

- ✓ (d) Write the applications of χ^2 - test.
- (e) Define Maximum Likelihood Estimate with example.
- (f) Define Student's t-distribution.

Section – C

(Long-answer Type Questions)

✓ 3. Answer any four questions of the following :

7×4 = 28

- ✓ (a) Discuss the three basic principles of Design.
- ✓ (b) What do you mean by Randomised Block Design (R. B. D.) ? Give its layout and analysis of variance.
- ✓ (c) Discuss two types of errors in testing of hypothesis.
- (d) Discuss the terms (a) Estimate (b) Unbiased estimate of a parameter. Show that sample mean is unbiased estimate of the population mean.
- ✓ (e) Define χ^2 distribution. Obtain its mean and variance.
- (f) Give the details description of Student's t-test and its applications.

GQ – 9/2 (500)

(4)

Voc(Sem-IV) — BCA
(GE – 4) Stat – II